EXPERIMENT 11: APPLICATION DEPLOYMENT AND MIGRATION

AIM

To establish, configure, and evaluate an automated multi-stage application deployment pipeline and lifecycle migration workflow within Zoho Creator for the 'Vehicle Service & Repair Request Management' enterprise system. The objective is to implement a continuous delivery architecture separating Development, Stage, and Production tiers, manage release versioning, and execute production publishing protocols.

APPLICATION CONTEXT & SYSTEM OVERVIEW

In professional low-code application development, updating operational database schemas, Deluge automation scripts, and user interface forms directly on active instances carries critical risks of workflow interruption and data corruption. Zoho Creator resolves this operational risk through Multi-Stage Environments. This capability establishes an isolated three-tier release pipeline: Development (for active coding and feature updates), Stage (for user acceptance testing and integration validation), and Production (for live customer and operational access). It guarantees structured change governance, seamless version rollouts, and consistent environment migration.

PROCEDURE

Step 1: Navigating to Administrative Applications Console

Access the central administration console of Zoho Creator under the registered developer tenant account. Inspect the deployed solutions portfolio, identifying the target system 'Vehicle Service & Repair Request Management' alongside existing workspace applications to verify repository availability.

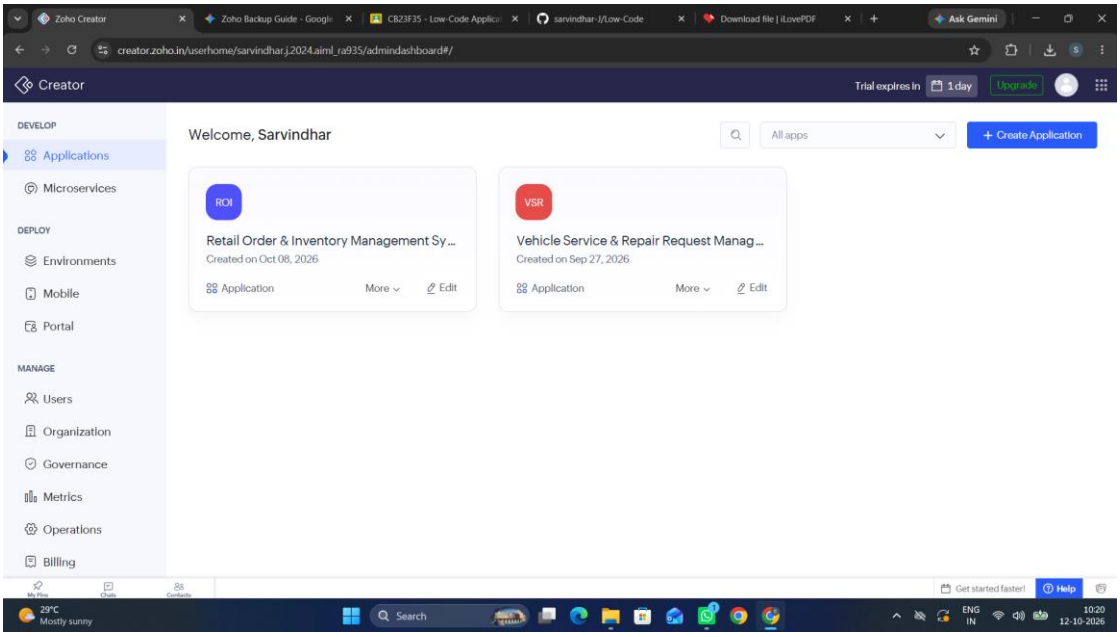


Figure 11.1: Administrative Applications Workbench and Workspace Overview

Step 2: Initializing Multi-Stage Deployment Environments

From the left navigation drawer under the 'DEPLOY' section, select the 'Environments' console. The multi-stage deployment workspace opens, presenting an unconfigured pipeline framework partitioned into Development, Stage, and Production deployment boundaries.

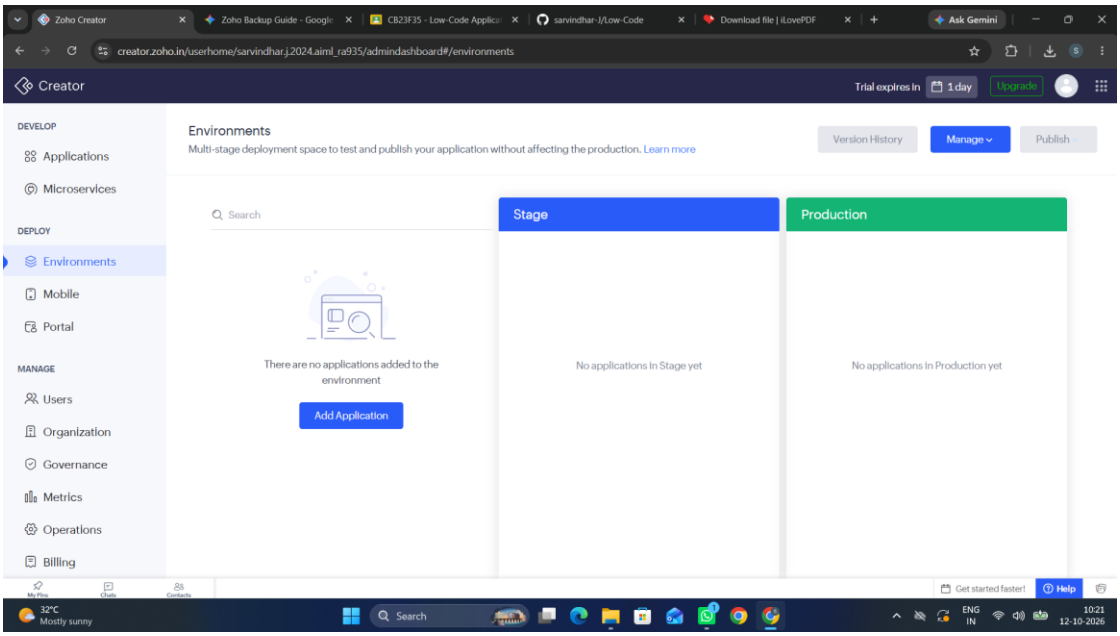


Figure 11.2: Multi-Stage Continuous Deployment Canvas Architecture

Step 3: Attaching Application and Provisioning Development Sandbox

Select the 'Add Application' action on the Environments canvas and choose the 'Vehicle Service & Repair Request Management' system. The platform initiates automated instance provisioning, generating a sandbox development fork while binding the baseline production instance to Version 1.0.

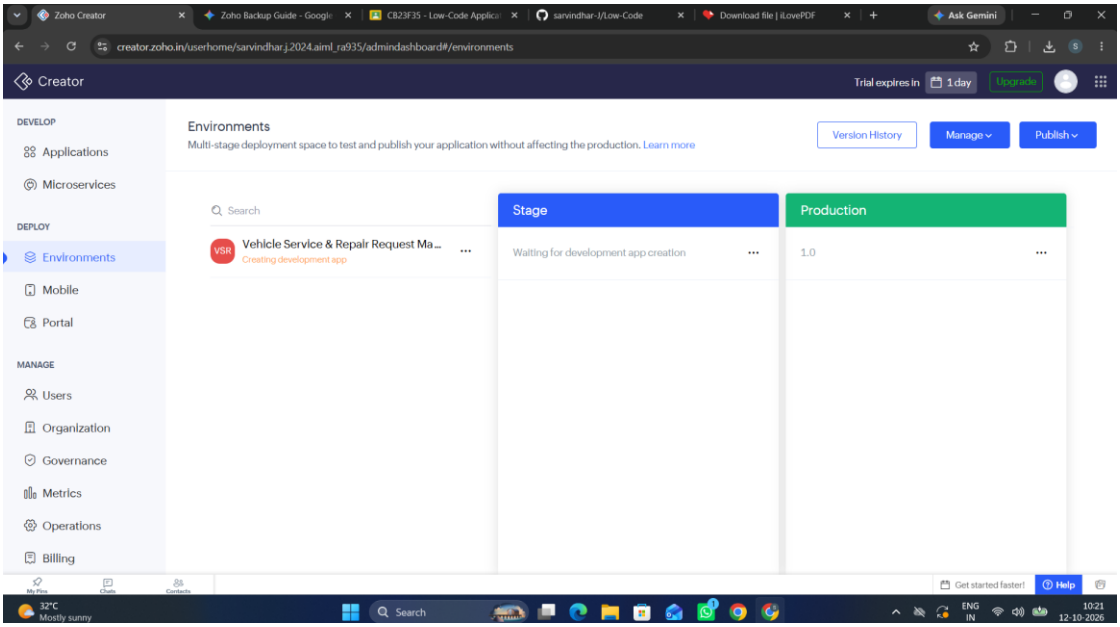


Figure 11.3: Pipeline Provisioning and Development Instance Creation

Step 4: Managing Environment Sandboxes and Allocation Quotas

Evaluate the deployment workspace status during instance binding. The administrative console verifies multi-tenant application boundaries, quotas, and release linkages between the local development sandbox and the upstream staging container.

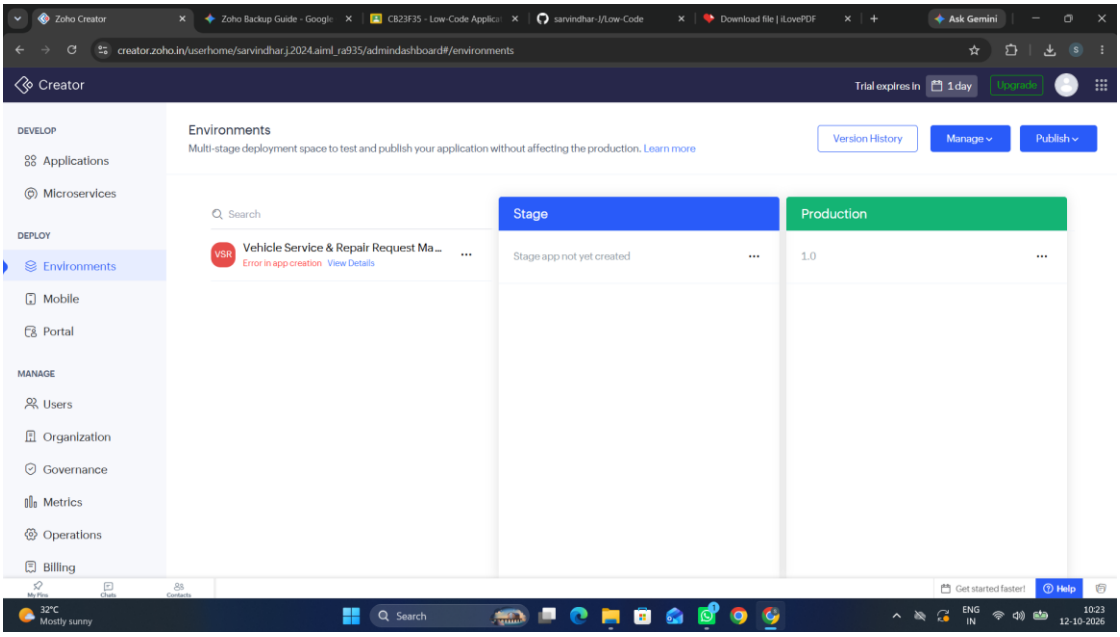


Figure 11.4: Deployment Environment Diagnostics and Pipeline Boundary Verification

Step 5: Production Deployment and Continuous Delivery Verification

Finalize the multi-tier release lifecycle pipeline. Validate the fully provisioned deployment state: the Development environment with 'Push to Stage' readiness, the Stage environment running release v1.0.1 (Draft) enabled with 'Test' and 'Promote' controls, and the Production environment serving active live traffic on Version 1.0.

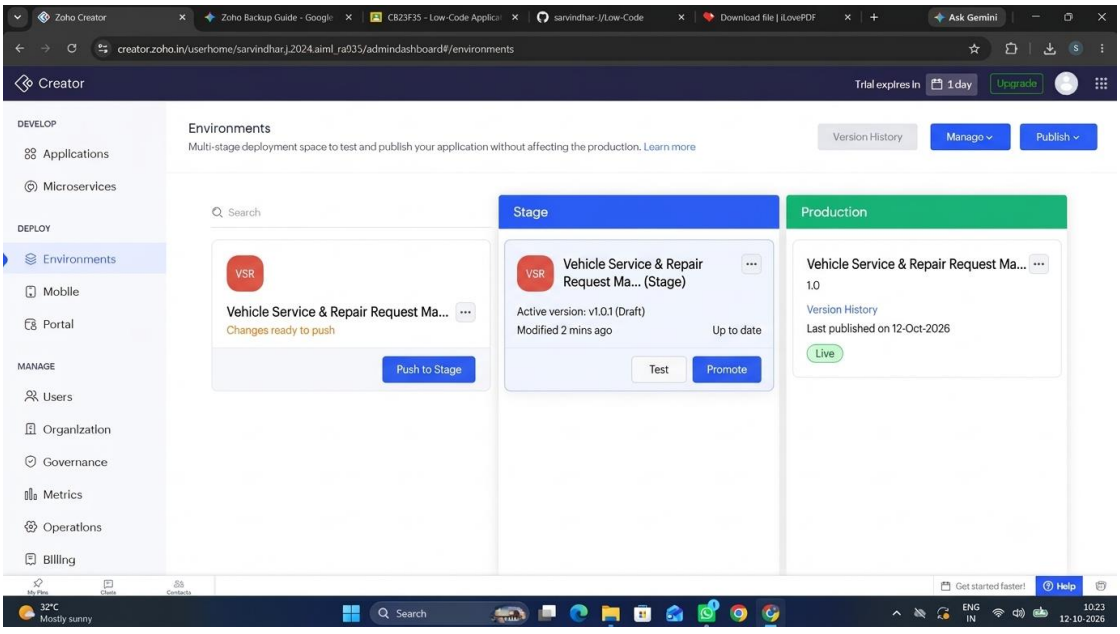


Figure 11.5: Production Deployment and Verified Multi-Stage Lifecycle Pipeline

RESULT

Thus, the design, setup, and verification of the multi-stage application deployment and migration pipeline for the 'Vehicle Service & Repair Request Management' enterprise system in Zoho Creator was successfully achieved. The three-tier release architecture spanning Development, Stage, and Production environments was fully established, validating automated change promotion, release version tagging, and reliable cloud lifecycle migration.